light of day looking at tracks left by antelope in the dust of the savannah – scrutinizing and deducting from the details of reality in order to pursue something which we can't see directly but can follow the traces of. In the awareness that we can always be wrong, and therefore ready at any moment to change direction if a new track appears; but knowing also that if we are good enough we will get it right and will find what we are seeking. This is the nature of science.

The confusion between these two diverse human activities – inventing stories and following traces in order to find something – is the origin of the incomprehension and distrust of science shown by a significant part of our contemporary culture. The separation is a subtle one: the antelope hunted at dawn is not far removed from the antelope deity in that night's storytelling.

The border is porous. Myths nourish science, and science nourishes myth. But the value of knowledge remains. If we find the antelope we can eat.

Our knowledge consequently reflects the world. It does this more or less well, but it reflects the world we inhabit. This communication between ourselves and the world is not what distinguishes us from the rest of nature. All things are continually interacting with each other, and in doing so each bears the traces of that with which it has interacted: and in this sense all things continuously exchange information about each other.

The information which one physical system has about another has nothing mental or subjective about it: it's only the connection that physics determines between the state of something and the state of something else. A raindrop contains information on the presence of a cloud in the sky; a ray of light contains information on the colour of the substance from which it came; a clock has information on the time of day; the wind carries information about an approaching storm; a cold virus has information of the vulnerability of my nose; the DNA in our cells contains all the information in our genetic code (on what makes me resemble my parents); and my brain teems with information accumulated from my experi-ence. The primal substance of our thoughts is an extremely rich gathering of information that's accumulated, exchanged and continually elaborated.

Even the thermostat on my central heating system 'senses' and 'knows' the ambient temperature in my home, has information on it and turns off when it is warm enough. So what then is the difference between the thermostat's and my own 'sensing' and 'knowing' that it's warm and deciding freely to turn off